9.1: Catabolic pathways yield energy by oxidizing organic fuels

Living cells require transfusions of energy from outside sources to perform their many tasks—for example, assembling polymers, pumping substances across membranes, moving, and reproducing. The outside source of energy is food, and the energy stored in the organic molecules of food ultimately comes from the sun. As shown in **Figure 9.1**, energy flows into an ecosystem as sunlight and exits as heat; in contrast, the chemical elements essential to life are recycled. Photosynthesis generates oxygen, as well as organic molecules used by the mitochondria of eukaryotes as fuel for cellular respiration. Respiration breaks this fuel down, using oxygen O_2 and generating ATP. The waste products of this type of respiration, carbon dioxide CO_2 and water H_2O, are the raw materials for photosynthesis.

Animation

Energy Flow and Chemical Recycling

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Let's consider how cells harvest the chemical energy stored in organic molecules and use it to generate ATP, the molecule that drives most cellular work. Metabolic pathways that release stored energy by breaking down complex molecules are called catabolic pathways (see Concept 8.1). Transfer of electrons from food molecules (like glucose) to other molecules plays a major role in these pathways. In this section, we consider these processes, which are central to cellular respiration.

Catabolic Pathways and Production of ATP

Organic compounds possess potential energy as a result of the arrangement of electrons in the bonds between their atoms. Compounds that can participate in exergonic reactions can act as fuels. Through the activity of enzymes (see Concept 8.4), a cell systematically degrades complex organic molecules that are rich in potential energy to simpler waste products that have less energy. Some of the energy taken out of chemical storage can be used to do work; the rest is dissipated as heat.

One catabolic process, **fermentation**, is a partial degradation of sugars or other organic fuel that occurs without the use of oxygen. However, the most efficient catabolic pathway is **aerobic respiration**, in which oxygen is consumed as a reactant along with the organic fuel (*aerobic* is from the Greek *aer*, air, and *bios*, life). The cells of most eukaryotic and many prokaryotic organisms can carry out aerobic respiration. Some prokaryotes use substances other than oxygen as reactants in a similar process that harvests chemical energy without oxygen; this process is called *anaerobic respiration* (the prefix *an*- means "without"). Technically, the term **cellular respiration** includes both aerobic and anaerobic processes. However, it originated as a synonym for aerobic respiration because of the relationship of that process to organismal respiration, in which an animal breathes in oxygen. Thus, *cellular respiration* is often used to refer to the aerobic process, a practice we follow in most of this chapter.

Although very different in mechanism, aerobic respiration is in principle similar to the combustion of gasoline in an automobile engine after oxygen is mixed with the fuel (hydrocarbons). Food provides the fuel for respiration, and the exhaust is carbon dioxide and water. The overall process can be summarized as follows:

$$\text{Organiccompounds} + \text{Oxygen} \rightarrow \text{Carbondioxide} + \text{Water} + \text{Energy}$$

Carbohydrates, fats, and proteins from food can all be processed and consumed as fuel. In animal diets, a major source of carbohydrates is starch, a storage polysaccharide that can be broken down into glucose $C_6H_{12}O_6$ subunits. Here, we will learn the steps of cellular respiration by tracking the degradation of the sugar glucose:

$$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + \text{Energy (ATP+heat)}$$

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Introduction to Cellular Respiration

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This breakdown of glucose is exergonic, having a free-energy change of −686 kcal (-2,870 kJ) per mole of glucose decomposed ($\Delta G = -686$ kcal/mol). Recall that a negative ΔG($\Delta G < 0$) indicates that the products of the chemical process store less energy than the reactants and that the reaction can happen spontaneously—in other words, without an input of energy (see Concept 8.2).

Catabolic pathways do not directly move flagella, pump solutes across membranes, polymerize monomers, or perform other cellular work. Catabolism is linked to work by a chemical drive shaft—ATP (see Concept 8.3). To keep working, the cell must regenerate its supply of ATP from ADP and Ⓟi (see Figure 8.12). To understand how cellular respiration accomplishes this, let's examine the fundamental chemical processes known as oxidation and reduction.

Redox Reactions: Oxidation and Reduction

How do the catabolic pathways that decompose glucose and other organic fuels yield energy? The answer is based on the transfer of electrons during the chemical reactions. The relocation of electrons releases energy stored in organic molecules, and this energy ultimately is used to synthesize ATP.

The Principle of Redox

In many chemical reactions, there is a transfer of one or more electrons e− from one reactant to another. These electron transfers are called oxidation-reduction reactions, or **redox reactions** for short. In a redox reaction, the loss of electrons from one substance is called **oxidation**, and the addition of electrons to another substance is known as **reduction**. (Note that *adding* electrons is called *reduction*; adding negatively charged electrons to an atom *reduces* the amount of positive charge of that atom.)

To take a simple, nonbiological example, consider the reaction between the elements sodium (Na) and chlorine (Cl) that forms table salt:

> N a plus C l yields N a superscript plus, plus C l superscript minus. N a becomes oxidized (loses electron) as N a superscript plus. C l becomes reduced (gains electron) as C l superscript minus.

We could generalize a redox reaction this way:

> X e minus plus Y yields X plus Y e minus. X e minus becomes oxidized as X. Y becomes reduced as Y e minus.

In the generalized reaction, substance Xe−, the electron donor, is called the **reducing agent**; it reduces Y, which accepts the donated electron. Substance Y, the electron acceptor, is the **oxidizing agent**; it oxidizes Xe− by removing its electron. Because an electron transfer requires both an electron donor and an acceptor, oxidation and reduction always go hand in hand.

Not all redox reactions involve the complete transfer of electrons from one substance to another; some change the degree of electron sharing in covalent bonds. Methane combustion, shown in **Figure 9.2**, is an example. The covalent electrons in methane are shared nearly equally between the bonded atoms because carbon and hydrogen have about the same affinity for valence electrons; they are about equally

electronegative (see Concept 2.3). But when methane reacts with O_2, forming CO_2, electrons end up shared less equally between the carbon atom and its new covalent partners, the oxygen atoms, which are very electronegative. In effect, the carbon atom has partially “lost” its shared electrons; thus, methane has been oxidized.

Figure 9.2

Methane combustion as an energy-yielding redox reaction.

C H 4 plus 2 O 2 goes to C O 2 plus Energy plus 2 H 2 O. C H 4 is oxidized into C O 2 and O 2 is reduced into H 2 O.

The reaction releases energy to the surroundings because the electrons lose potential energy when they end up being shared unequally, spending more time near electronegative atoms such as oxygen.

Electron sharing in covalent bonds is shown for each reactant and product. In methane, the reducing agent, carbon shares a pair of electrons equally in four covalent bonds with each of the four hydrogen atoms. In oxygen, the oxidizing agent, two pairs of electrons are equally shared in a covalent double bond between the two oxygen atoms. In carbon dioxide, carbon shares two pairs of electrons with each of two oxygen atoms, but the electrons are not shared equally and are pulled towards both oxygen atoms. In water, oxygen shares a pair of electrons with each of two hydrogen atoms, but the electrons are not shared equally and are pulled towards oxygen.

VISUAL SKILLS *Is the carbon atom oxidized or reduced during this reaction? Explain.*

*For suggested answer, see*Appendix A.**

Animation

Redox Reactions

image

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Now let’s examine the fate of the reactant O_2. The two atoms of O_2 share their electrons equally. But after the reaction with methane, when each O atom is bonded to two H atoms in H_2O, the electrons of those covalent bonds spend more time near the oxygen (see Figure 9.2). In effect, each O atom has partially “gained” electrons, so the oxygen molecule O_2 has been reduced. Because the O atom is so electronegative, O_2 is one of the most powerful of all oxidizing agents.

Energy must be added to pull an electron away from an atom, just as energy is required to push a ball uphill. The more electronegative the atom (the stronger its pull on electrons), the more energy is required to take an electron away from it. An electron loses potential energy when it shifts from a less electronegative atom toward a more electronegative one, just as a ball loses potential energy when it rolls downhill. A redox reaction that moves electrons closer to an O atom, such as the burning (oxidation) of methane, therefore releases chemical energy that can be put to work.

Oxidation of Organic Fuel Molecules During Cellular Respiration

The oxidation of methane by O_2 is the main combustion reaction that occurs at the burner of a gas stove. The combustion of gasoline in an automobile engine is also a redox reaction; the energy released pushes the pistons. But the energy-yielding redox process of greatest interest to biologists is respiration: the oxidation of glucose and other molecules in food. Examine again the summary equation for cellular respiration, but this time think of it as a redox process:

> $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$. $C_6H_{12}O_6$ becomes oxidized as $6CO_2$. $6O_2$ becomes reduced as $6H_2O$.

As in the combustion of methane or gasoline, the fuel (glucose) is oxidized and O_2 is reduced. The electrons lose potential energy along the way, and energy is released.

In general, organic molecules that have an abundance of hydrogen are excellent fuels because their bonds are a source of “hilltop” electrons, whose energy may be released as these electrons “fall” down an energy gradient during their transfer to oxygen. The summary equation for respiration indicates that hydrogen is transferred from glucose to the O atoms in O_2. But the important point, not visible in the summary equation, is that the energy state of the electron changes as hydrogen (with its electron) is transferred to oxygen. In respiration, the oxidation of glucose transfers electrons to a lower energy state, liberating energy that becomes available for ATP synthesis. So, in general, we see fuels with multiple C—H bonds oxidized into products with multiple C—O bonds.

The main energy-yielding foods—carbohydrates and fats—are reservoirs of electrons associated with hydrogen, often in the form of C—H bonds. Only the barrier of activation energy holds back the flood of electrons to a lower energy state (see Figure 8.13). Without this barrier, a food substance like glucose would combine almost instantaneously with O_2. If we supply the activation energy by igniting glucose, it burns in air, releasing 686 kcal (2,870 kJ) of heat per mole of glucose (about 180 g). Body temperature is not high enough to initiate burning, of course. Instead, if you swallow some glucose, enzymes in your cells will lower the barrier of activation energy, allowing the sugar to be oxidized in a series of steps.

Stepwise Energy Harvest via NAD+ and the Electron Transport Chain

If energy is released from a fuel all at once, it cannot be harnessed efficiently for constructive work. For example, if a gasoline tank explodes, it cannot drive a car very far. Cellular respiration does not oxidize glucose (or any other organic fuel) in a single explosive step either. Rather, glucose is broken down in a series of steps, each one catalyzed by an enzyme. At key steps, electrons are stripped from the glucose. As is often the case in oxidation reactions, each electron travels with a proton—thus, as a hydrogen atom. The hydrogen atoms are not transferred directly to O2, but instead are usually passed first to an electron carrier, a coenzyme called nicotinamide adenine dinucleotide, a derivative of the vitamin niacin. This coenzyme is well suited as an electron carrier because it can cycle easily between its oxidized form, ****__NAD plus__, and its reduced form, **NADH**. As an electron acceptor, NAD+ functions as an oxidizing agent during respiration.

How does NAD+ trap electrons from glucose and the other organic molecules in food? Enzymes called dehydrogenases remove a pair of hydrogen atoms (2 electrons and 2 protons) from the substrate (glucose, in the preceding example), thereby oxidizing it. The enzyme delivers the 2 electrons along with 1 proton to its coenzyme, NAD+, forming NADH **(Figure 9.3)**. The other proton is released as a hydrogen ion H+ into the surrounding solution:

A reaction is given.

H, single bond, C, single bond, O H, with single bonds below and above C, is added to N A D plus. In the presence of dehydrogenase, it yields C, double bond O, where C has single bonds below and above, plus N A D H plus H plus.

Figure 9.3

NAD+ as an electron shuttle.

N A D plus as an electron shuttle.

Reversible reaction with enzyme dehydrogenase: to the right, reduction of N A D plus; to the left, oxidation of N A D H. Reaction is N A D plus, plus 2 H (from food) equilibrium arrow to N A D H plus H plus. The 2 H (from food) represents 2 electrons plus 2 H plus from food. One H plus, plus 2 e minus reduces N A D plus to N A D H, leaving one H plus. The structure of N A D plus is two nucleotides joined at their two phosphate groups. Nicotinamide consists of an amino group attached to a carbonyl group attached to a nitrogen-containing ring that has one more hydrogen attached in the reduced form (N A D H) than it has in the oxidized form (N A D plus).

The full name for NAD+, nicotinamide adenine dinucleotide, describes its structure—the molecule consists of two nucleotides joined together at their phosphate groups (shown in yellow). (Nicotinamide is a nitrogenous base, although not one that is present in DNA or RNA; see Figure 5.23.) The enzymatic transfer

of 2 electrons and 1 proton (H+) from an organic molecule in food to NAD+reduces the NAD+to NADH: Most of the electrons removed from food are transferred initially to NAD+, forming NADH.

VISUAL SKILLS *Describe the structural differences between the oxidized form and the reduced form of nicotinamide.*

*For suggested answer, see*Appendix A.**

By receiving 2 negatively charged electrons but only 1 positively charged proton, the nicotinamide portion of NAD+ has its charge neutralized when NAD+ is reduced to NADH. The name NADH shows the hydrogen that has been received in the reaction. NAD+ is the most versatile electron acceptor in cellular respiration and functions in several of the redox steps during the breakdown of glucose.

Electrons lose very little of their potential energy when they are transferred from glucose to NAD+. Each NADH molecule formed during respiration represents stored energy that can be tapped to make ATP when the electrons complete their "fall" down an energy gradient from NADH to O2.

How do electrons that are extracted from glucose and stored as potential energy in NADH finally reach oxygen? It will help to compare the redox chemistry of cellular respiration to a much simpler reaction: the reaction between hydrogen and oxygen to form water **(Figure 9.4a)**. Mix H2 and O2, provide a spark for activation energy, and the gases combine explosively. In fact, combustion of liquid H2 and O2 is harnessed to help power the rocket engines that boost satellites into orbit and launch spacecraft. The explosion represents a release of energy as the electrons of hydrogen "fall" closer to the electronegative oxygen atoms. Cellular respiration also brings hydrogen and oxygen together to form water, but there are two important differences. First, in cellular respiration, the hydrogen that reacts with oxygen is derived from organic molecules rather than H2. Second, instead of occurring in one explosive reaction, respiration uses an electron transport chain to break the fall of electrons to oxygen into several energy-releasing steps **(Figure 9.4b)**. An **electron transport chain** consists of a number of molecules, mostly proteins, built into the inner membrane of the mitochondria of eukaryotic cells (and the plasma membrane of respiring prokaryotes). Electrons removed from glucose are shuttled by NADH to the "top," higher-energy end of the chain. At the "bottom," lower-energy end, O2 captures these electrons along with hydrogen nuclei H+, forming water. (Anaerobically respiring prokaryotes have an electron acceptor at the end of the chain that is different from O2.)

Figure 9.4

An introduction to electron transport chains.

An introduction to the electron transport chains.

Part a. Uncontrolled reaction. The one-step exergonic reaction of hydrogen with oxygen to form water releases a large amount of energy in the form of heat and light: an explosion. 2 H plus one-half O 2 yields one H 2 O. Free Energy (G) increases in opposition to the one-step reaction, resulting in an explosive release of heat and light energy.

Part b. Cellular respiration. The same reaction occurs in stages: An electron transport chain breaks the fall of electrons in this reaction into smaller steps and stores some of the released energy in a form that can be used to make ATP. (The rest of the energy is released as heat.) Free energy (G) increases in opposition to the stepwise progress of the reaction H 2 plus one-half O 2 goes to H 2 0, controlling the release of energy for the synthesis of A T P. 2 electrons coming from food via N A D H are passed through a series of controlled steps in the electron transport chain. At the bottom of the chain, the 2 electrons and 2 H plus combine with one-half 0 2 to form water. In the process, A T P is synthesized.

Electron transfer from NADH to oxygen is an exergonic reaction with a free-energy change of −53 kcal/mol (−222 kJ/mol). Instead of this energy being released and wasted in a single explosive step, electrons cascade down the chain from one carrier molecule to the next in a series of redox reactions, losing a small amount of energy with each step until they finally reach oxygen, the terminal electron acceptor, which has a very great affinity for electrons. Each “downhill” carrier has a greater affinity for electrons than, and is thus capable of accepting electrons from (oxidizing), its “uphill” neighbor, with O2 at the bottom of the chain. Therefore, the electrons transferred from glucose to NAD+, reducing it to NADH, fall down an energy gradient in the electron transport chain to a far more stable location in an electronegative oxygen atom from O2. Put another way, O2 pulls electrons down the chain in an energy-yielding tumble analogous to gravity pulling objects downhill.

In summary, during cellular respiration, most electrons travel the following “downhill” route: glucose→NADH→electron transport chain→oxygen. Later in this chapter, you will learn more about how the cell uses the energy released from this exergonic electron fall to regenerate its supply of ATP. For now, having covered the basic redox mechanisms of cellular respiration, let’s look at the entire process by which energy is harvested from organic fuels.

The Stages of Cellular Respiration: *A Preview*

The harvesting of energy from glucose by cellular respiration is a cumulative function of three metabolic stages. We list them here along with a color-coding scheme we will use throughout the chapter to help you keep track of the big picture:

Three stages of cellular respiration.

One. Glycolysis (color-coded blue throughout the chapter).

Two. Pyruvate oxidation (light orange) and the citric acid cycle (dark orange). Three.

Oxidative phosphorylation: electron transport and chemiosmosis (purple).

Biochemists usually reserve the term *cellular respiration* for stages 2 and 3 together. In this text, however, we include glycolysis as a part of cellular respiration because most respiring cells deriving energy from glucose use glycolysis to produce the starting material for the citric acid cycle.

As diagrammed in **Figure 9.5**, glycolysis and then pyruvate oxidation and the citric acid cycle are the catabolic pathways that break down glucose and other organic fuels. **Glycolysis**, which occurs in the cytosol, begins the degradation process by breaking glucose into two molecules of a compound called pyruvate. In eukaryotes, pyruvate enters the mitochondrion and is oxidized to a compound called acetyl CoA, which enters the **citric acid cycle**. There, the breakdown of glucose to carbon dioxide is completed. (In prokaryotes, these processes take place in the cytosol.) Thus, the carbon dioxide produced by respiration represents fragments of oxidized organic molecules.

Figure 9.5

An overview of cellular respiration.

An overview of cellular respiration.

Glycolysis occurs in the cytosol. In this process, glucose yields pyruvate. A T P is produced with substrate-level phosphorylation. Electrons are carried via N A D H into the mitochondrion. The pyruvate goes into the mitochondrion and pyruvate oxidation occurs with acetyl Co A. That goes into the citric acid cycle, which produces A T P with substrate-level phosphorylation. Electrons from the cycle are carried via N A D H and F A D H 2. These electrons and the electrons from glycolysis go into oxidative phosphorylation (electron transport and chemiosmosis). This produces A T P with oxidative phosphorylation.

During glycolysis, each glucose molecule is broken down into two molecules of pyruvate. In eukaryotic cells, as shown here, the pyruvate enters the mitochondrion. There it is oxidized to acetyl CoA, which will be further oxidized to CO_2 in the citric acid cycle. The electron carriers NADH and $FADH_2$ transfer electrons derived from glucose to electron transport chains. During oxidative phosphorylation, electron transport chains convert the chemical energy to a form used for ATP synthesis in the process called chemiosmosis. (During earlier steps of cellular respiration, a few molecules of ATP are synthesized in a process called substrate-level phosphorylation.) To visualize these processes in their cellular context, see Figure 6.32b.

Animation

Overview of Cellular Respiration

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Some of the steps of glycolysis and the citric acid cycle are redox reactions in which dehydrogenases transfer electrons from substrates to NAD+ or the related electron carrier FAD, forming NADH or FADH2. (You'll learn more about FAD and FADH2 later.) In the third stage of respiration, the electron transport chain accepts electrons from NADH or FADH2 generated during the first two stages and passes these electrons down the chain. At the end of the chain, the electrons are combined with molecular oxygen O2 and hydrogen ions H+, forming water (see Figure 9.4b). The energy released at each step of the chain is stored in a form the mitochondrion (or prokaryotic cell) can use to make ATP from ADP. This mode of ATP synthesis is called **oxidative phosphorylation** because it is powered by the redox reactions of the electron transport chain.

In eukaryotic cells, the inner membrane of the mitochondrion is the site of electron transport and another process called *chemiosmosis*, together making up oxidative phosphorylation. (In prokaryotes, these processes take place in the plasma membrane.) Oxidative phosphorylation accounts for almost 90% of the ATP generated by respiration. A smaller amount of ATP is formed directly in a few reactions of glycolysis and the citric acid cycle by a mechanism called **substrate-level phosphorylation(Figure 9.6)**. This mode of ATP synthesis occurs when an enzyme transfers a phosphate group from a substrate molecule to ADP, rather than adding an inorganic phosphate to ADP as in oxidative phosphorylation. "Substrate molecule" here refers to an organic molecule generated as an intermediate during the catabolism of glucose. You'll see examples of substrate-level phosphorylation later in the chapter, in both glycolysis and the citric acid cycle.

Figure 9.6

Substrate-level phosphorylation.

Substrate with attached phosphate group binds to the active site of an enzyme, which transfers phosphate to ADP, forming ATP. Substrate is then released by the enzyme as a product.

Some ATP is made by direct transfer of a phosphate group from an organic substrate to ADP by an enzyme. (For examples in glycolysis, see Figure 9.8, steps 7 and 10.)

MAKE CONNECTIONS *Review*Figure 8.9*. In the reaction shown above, is the potential energy higher for the reactants or for the products? Explain.*

*For suggested answer, see*Appendix A*.*

You can think of the whole process this way: When you withdraw a relatively large sum of money from an ATM, it is not delivered to you in a single bill of a large denomination. Instead, the machine dispenses a number of smaller-denomination bills that you can spend more easily. This is analogous to ATP production during cellular respiration. For each molecule of glucose degraded to CO_2 and H_2O by respiration, the cell makes up to about 32 molecules of ATP, each with 7.3 kcal/mol of free energy. Respiration cashes in the large denomination of energy banked in a single molecule of glucose (686 kcal/mol under standard conditions) for the small change of many molecules of ATP, which is more practical for the cell to spend on its work.

This preview has introduced you to how glycolysis, the citric acid cycle, and oxidative phosphorylation fit into the process of cellular respiration so you can keep the big picture in mind as you take a closer look at each of these three stages of respiration. As you read about the chemical reactions, remember that each reaction is catalyzed by a specific enzyme, some of which are shown in Figure 6.32b.